Local models (Whisper) — participant handout

BrainHack 2026 Workshop — Speech Recognition and Generation
Module 4 (30 min) · Instructors: Franciszek Kubala, Mateusz Poszelężny — KN AIMed AGH

> This material accompanies the hands-on part. You will find concepts, comparisons and links here. The code is in the `Local_ASR_models.ipynb` notebook — you do not have to write it yourself.

1. What is a "local model" (Whisper)?

Whisper is a speech recognition model that you can run **on your own machine**, on your own computer. In a Google Colab session the same model runs, but on a Google machine — so that is already the cloud.

Analogy: an interpreter sitting in the room with you, rather than reached by phone at a translation agency.

Key point: Whisper does not send audio to an ASR provider's API. But **where the file lives depends on where you run it:**

- **Your own computer** — the audio stays with you.
- **Google Colab** — the model runs without an ASR API, but the file goes to a Google machine. Colab is the cloud, not "local".
- **API (ElevenLabs)** — the audio goes to the ASR provider's servers.

2. Cloud vs local

Question	Cloud (API, e.g. ElevenLabs)	Colab (Whisper in the Google cloud)	Own computer (Whisper)
Where the audio goes	To the ASR provider's servers	To a Google machine, without an ASR API	Stays on your computer
Retention	Depends on the provider (check the policy)	Depends on Google and the session; files disappear from the session when it ends	You decide what to keep
Training models on your data	Possible (opt-out)	Check Google's terms	Not applicable
Quality	Often very good	Depends on the model and resources	Depends on the model and resources
Time and convenience	Fast, no installation	No installation, in the browser	Requires installation
Cost	Limits/payments	Free limits	Free (depends on hardware)
Control over data	Limited	Limited	Full

Takeaway: the choice depends on what matters more — convenience/quality or privacy/control. If the recording is sensitive, only your own computer remains.

3. Privacy and consent (psychology)

Checklist before recording a session/interview:

- [ ] Patient/client **consent** (informed, voluntary, revocable).
- [ ] A clearly defined recording **purpose**.
- [ ] **Where** the recording and transcript will go (cloud or local).
- [ ] **Who** has access and for how long (retention).
- [ ] **Anonymization** / pseudonymization in notes.
- [ ] **Professional confidentiality** and data minimization.

Rule: if the recording is sensitive — prefer a local model and do not send audio to the cloud without a legal basis.

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4. What it looks like in practice (Colab) — without code

1. Open the `Local_ASR_models.ipynb` notebook in Google Colab.
2. Run the installation cell (once).
3. Upload your own recording (~1 min).
4. Run the transcription (the model runs in the Colab session).
5. Save the result to a file and download it to your disk.

Where the code is: in the notebook — you just run the cells (`Shift+Enter`). You do not have to write code yourself.

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5. Saving the transcript

- **txt** — plain text. For archives, qualitative analysis, quotations.
- **SRT** — subtitles with timestamps (format `HH:MM:SS,mmm`). For video, for listening back to specific fragments.

Why save: archiving, verification, quoting, analysis.

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6. Quality: WER and CER

- **WER** (Word Error Rate) = word errors / all reference words.
- **CER** (Character Error Rate) = the same at the character level.
- **Interpretation:** lower is better; 0% = perfect.
- **Typically:** a good transcription of clean speech is ~5–10% WER.
- **What affects it:** noise, accents, jargon, overlapping speech, recording quality.

How we compute it in the workshop: we compare a manual transcript (reference) with the model's output (hypothesis) — the `jiwer` library.

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7. Diarization — in brief

- **What it is:** "who speaks and when" — splitting a recording into segments assigned to speakers.
- **Locally:** e.g. WhisperX or pyannote (requires more resources and often a Hugging Face token).
- **Today:** we show the idea; details and experiments — on your own (notebook, docs).

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8. When cloud, when local?

Choose **local** when:

- the recording is sensitive (sessions, interviews, patient data),
- you need full control over the data,
- you have no legal basis for sending audio to the cloud.

Choose **cloud** when:

- convenience and speed matter,
- quality must be top (e.g. a difficult accent, jargon),
- you have consent and the legal requirements are met.

9. Glossary

Term	Definition
Local model	A model running on your own computer, without sending audio anywhere
Cloud	Processing on the provider's servers
Retention	How long data is stored
Training opt-out	Turning off the use of your data for training models
WER	Word Error Rate — errors at the word level
CER	Character Error Rate — errors at the character level
SRT	A subtitle format with timestamps
Colab	Google Colab — a notebook in the Google cloud
HF token	A Hugging Face token (access to models)
Diarization	Splitting a recording into segments assigned to speakers

10. Useful sources

- faster-whisper: https://github.com/SYSTRAN/faster-whisper
- openai-whisper: https://github.com/openai/whisper
- jiwer (WER/CER): https://github.com/jitsi/jiwer
- WhisperX: https://github.com/m-bain/whisperX
- pyannote: https://github.com/pyannote/pyannote-audio
- Google Colab: https://colab.research.google.com
- Whisper (paper): https://arxiv.org/abs/2212.04356

11. What's next

- The `Local_ASR_models.ipynb` notebook — run it in Colab on your own recording.
- Module 5 — summary and questions.

Materials: BrainHack2026_ASR_Workshop repo · AIMed AGH